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Master's Degree In Aerospace Engineering
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[EXAMPLE] Adaptive wall modelling approach in Large Eddy Simulation with the Immersed Boundary Method.

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Chapter 1

Introduction

1.1 Computational Fluid Dynamics

Computational Fluid Dynamics (CFD) is the study of fluids through a numerical computer-based approach, using algorithms to find solutions to the equations that describe the fluid behavior. The information produced have mainly two purposes: in the field of research, it allows a better understanding of the physical mechanisms governing the fluid motion, and the consequent modelling and control of such mechanisms. In the engineering analyses, fluid simulation is used as a prediction tool, especially during the product design phase, to assess flow characteristics and physical parameters, such as lift and drag forces generated, propagation of acoustic waves or convective heat transfer.

As all mathematical models, the definition of the level of approximation have a primary importance also for fluid simulation. This choice concern various aspects of the description of the phenomena. Firstly the scale of reality considered: the reality can be described at an atomic, particle or micro- and macroscopic level. The last one is that used by classical mechanics, in particular continuum mechanics, that is the case considered in CFD.

The second aspect to be defined in term of level of approximation is the accuracy of dynamic description. It consist on the choice of which forces should be considered exerted on the system, based on their relevance. The most general case is represented by the Navier-Stokes equations complemented by constitutive laws (empirical laws characterizing the fluid and state laws). This case can be simplified, obtaining for example the Euler equations, for the description of inviscid flows.

Another aspect regards the scale separation problem, in particular the spatial and temporal resolution of choice. To obtain a description of maximum quality, all the space-time scales of the solution must be taken into account while solving the unsteady Navier-Stokes

equations. As will be discussed later, it turns out that such a solution criterion is extremely restrictive for multi-scale phenomena, like in the case of turbulent flows.

1.2 Turbulent flows

In most of the engineering applications, flows are prevalently turbulent. Generally this is the case of fluid processed by pumps, compressors or flows around cars, trains, airplanes and many more situations in the everyday surrounding.

Such flows are characterized by some specific properties. A key feature is the multi-scale nature of the vortical structures, whose dimensions span 6 to 8 orders of magnitude. Despite the deterministic nature (they can be described accurately by the Navier-Stokes equations), the strong dependence and sensitivity to initial conditions make their behavior chaotic. Furthermore, turbulent flows are intrinsically three-dimensional and unsteady, they show an high diffusivity (mixing of momentum, energy and species) and also an increased wall friction when compared to laminar flows.

The flow regime is described by the Reynolds number, a non-dimensional value representing the ratio between inertial and viscous forces, and defined as:

$$Re = \frac{\rho\, U_0\, L_0}{\mu} = \frac{U_0\, L_0}{\nu} \tag{1.1}$$

where U_0 and L_0 are the characteristic velocity and length scale of the flow. It is observed that, for higher values of Re, the magnitude of velocity gradients is progressively more intense, and the scale of smaller vortices decreases.

Considering the chaotic nature of the turbulence, a deterministic and instantaneous description of the phenomena is not always the most efficient option. Consequently, the analysis is typically focused on a statistical approach, considering averaged fields. It can be observed how those averaged fields resemble qualitatively the laminar behavior, and their dependence on the Reynolds number diminishes significantly.

1.2.1 Reynolds-Averaged Navier-Stokes equations

To analytically formalize the idea of considering the average fields and the turbulent one separately, the Reynolds averaging operator and the Reynolds decomposition are here introduced. The former consist in a ensemble average over a number of realizations. In more practical terms this implies repeating the same experiment N times, under identical conditions, and averaging the flow variables at the exact same location and time instant across all the independent measures. Considering to apply the Reynolds operator for the

velocity component u:

$$\langle u(\boldsymbol{x},t)\rangle = U(\boldsymbol{x},t) = \frac{1}{N}\sum_{k=1}^{N\gg 1} u_{(k)}(\boldsymbol{x},t) \tag{1.2}$$

Instead for what concerns the Reynolds decomposition, it can be defined as:

$$u(\boldsymbol{x},t) = U(\boldsymbol{x},t) + u'(\boldsymbol{x},t) \tag{1.3}$$

where u' is the fluctuation component and U is the averaged quantity.

By substituting in the Navier-Stokes equations the Reynolds decomposition and applying the Reynolds operator the Reynolds-Averaged Navier-Stokes (RANS) equations are obtained:

$$\begin{cases} \boldsymbol{\nabla}\cdot\boldsymbol{U} = 0 \\ \dfrac{D\boldsymbol{U}}{Dt} = -\dfrac{\boldsymbol{\nabla}P}{\rho} + \boldsymbol{\nabla}\cdot(2\nu\boldsymbol{E}) - \boldsymbol{\nabla}\cdot\langle\boldsymbol{u}'\boldsymbol{u}'\rangle \end{cases} \tag{1.4}$$

where the term $\langle\boldsymbol{u}'\boldsymbol{u}'\rangle$ is the Reynolds stress tensor, and its modelling represents the closure problem of the RANS equations.

1.2.2 Energy cascade

The RANS equations introduced above, allow demonstrating some fundamental mechanisms governing the energy dynamics within turbulent flows. Omitting for brevity the detailed algebraic steps, the mean and turbulent flow kinetic energies can be defined respectively as:

$$\mathcal{K} = \frac{1}{2}\boldsymbol{U}\cdot\boldsymbol{U} = \frac{1}{2}U_i U_i \qquad k = \frac{1}{2}\langle\boldsymbol{u}'\cdot\boldsymbol{u}'\rangle = \frac{1}{2}\langle u_i' u_i'\rangle \tag{1.5}$$

By manipulating the momentum equation, it is possible to derive a formulation for both the time derivative of the mean kinetic energy and turbulent kinetic energy terms:

$$\frac{D\mathcal{K}}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{P}{\rho}U_j + 2\nu\, E_{ij}\, U_i - \langle u_i'u_j'\rangle\, U_i\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_{\mathcal{K}}} \underbrace{-2\nu\, E_{ij}E_{ij}}_{-\mathcal{D}_{\mathcal{K}}} + \underbrace{\left(\langle u_i'u_j'\rangle\frac{\partial U_i}{\partial x_j}\right)}_{-\mathcal{P}} \tag{1.6}$$

$$\frac{Dk}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{\langle p'u_j'\rangle}{\rho} + 2\nu\,\langle e_{ij}'u_i'\rangle - \frac{\langle u_i'u_i'u_j'\rangle}{2}\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_k} \underbrace{-2\nu\,\langle e_{ij}'e_{ij}'\rangle}_{-\mathcal{E}_k} + \underbrace{\left(-\langle u_i'u_j'\rangle\frac{\partial U_i}{\partial x_j}\right)}_{\mathcal{P}} \tag{1.7}$$

where $\boldsymbol{F}_{\mathcal{K}}$ and $\boldsymbol{F}_k$ are the fluxes of kinetic energy (spatial transport terms), $\mathcal{D}_{\mathcal{K}}$ and $\mathcal{E}_k$ are the dissipation terms respectively for the mean and turbulent flows and $\mathcal{P}$ is the turbulent

energy production. In a more compact form, the two equation can be written as:

$$\frac{D\mathcal{K}}{Dt} = \boldsymbol{\nabla} \cdot \boldsymbol{F}_{\mathcal{K}} - \mathcal{D}_{\mathcal{K}} - \mathcal{P} \qquad \frac{Dk}{Dt} = \boldsymbol{\nabla} \cdot \boldsymbol{F}_k - \mathcal{E}_k + \mathcal{P} \tag{1.8}$$

Following the approach described by Pope [1], the equations can be integrated over a sufficiently large control volume Ω, surrounding a free shear flow such as a turbulent jet ($Re \gg 1$), in stationary conditions. This perspective, and what result from the integration, allows to identify which component are the dominant in the energy dynamics.

$$\int_\Omega \frac{\partial \mathcal{K}}{\partial t}\, d\Omega = \mathbb{C}_{\mathcal{K}} + \mathbb{F}_{\mathcal{K}} - \mathbb{D}_{\mathcal{K}} - \mathbb{P} \implies \mathbb{C}_{\mathcal{K}} \approx \mathbb{P} \tag{1.9}$$

$$\int_\Omega \frac{\partial k}{\partial t}\, d\Omega = \mathbb{C}_k + \mathbb{F}_k - \mathcal{E}_k + \mathbb{P} \implies \mathbb{P} \approx \mathcal{E}_k \tag{1.10}$$

The analysis of the dominant terms in the energy balance equations reveals how the input of energy, associated with convection ($\mathbb{C}_{\mathcal{K}}$) and its viscous dissipation ($\mathcal{E}_k$) occur at different scales. The apparent decoupling between energy input and output is better described by Richardson's theory.

Richardson's theory

The first concept in Richardson's view of the energy cascade is that the turbulence can be considered to be composed of eddies of different sizes [1].

Considering the eddies with the largest sizes, their characteristic dimensions can be evaluated as: $\ell \approx \ell_0 \approx L_0$ and $u \approx u_0 \approx u'_{rms} = \sqrt{(2/3) \cdot k} \approx (0.1 - 0.3)\, U_0$ (where the last relation comes from observation), thus their Reynolds number:

$$Re_0 = \frac{u_0\, \ell_0}{\nu} \approx \frac{U_0\, L_0}{\nu} \gg 1 \tag{1.11}$$

this suggests how at these scales the direct effect of molecular viscosity is negligible.

Thus, in order to achieve dissipation, unstable large eddies transfer energy to smaller ones. This process continues until the scale $\ell = \eta$ is reached, at this level the associated Reynolds number becomes sufficiently small, or in other words the molecular viscosity begins to be effective in energy dissipation.

The energy transferred to smaller scales can be evaluated as:

$$\frac{k_0}{\tau_0} \approx \frac{u_0^2}{(\ell_0/u_0)} = \frac{u_0^3}{\ell_0} \propto \frac{U_0^3}{L_0} \propto \mathcal{E}_k \tag{1.12}$$

the scaling of $\mathcal{E}_k$ with U_0^3/L_0, independently from ν (at the high Reynolds numbers considered) proves to be consistent with the experimental results for free shear flows [1].

Bibliography

[1] Stephen B. Pope. *Turbulent Flows*. Cambridge University Press, 2000.